

Weight Illusion from a "1st+3rd" Person Perspective: Integrating the Strengths of the 1st and 3rd Person Perspectives

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Abstract

The purpose of this study is to obtain a greater weight illusion in VR. In VR, the Proteus effect has been shown to have the potential to affect human perception. For example, it has been suggested that there is a certain correlation between the impression of the avatar's strength and perceived weight. In addition, a high sense of agency and direct visibility of the avatar, which are related to the features of the 1st and 3rd person perspectives respectively, may enhance the impression of the avatar's strength. To obtain a greater weight illusion, we have previously proposed the "1st+3rd" person perspective, which combines the advantages of the 1st and 3rd person perspectives. In this paper, with a new weight comparison task, we investigate the influence of the impression of the avatar's strength on the weight illusion in detail.

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1 Introduction

In recent years, within the field of virtual reality (VR) psychology, the Proteus effect where the appearance of an avatar influences the user's psychology, attitudes, and behavior has received considerable attention [Yee and Bailenson 2007]. This effect has also been shown to influence human perception. As a starting point for elucidating this, the effect of embodying VR avatars with different muscle masses on users' perception of weight was verified [Tunoda et al. 2020]. Their results revealed a tendency for users to perceive objects as lighter when embodying more muscular avatars. In addition, when viewing real-time images captured from a 3rd person perspective behind the user through a head-mounted display (HMD), it was reported that a perspective in which part of the user's body was made semi-transparent was preferred over the normal 3rd person perspective or the 1st person perspective for various tasks (such as opening doors) [Salamin et al. 2008].

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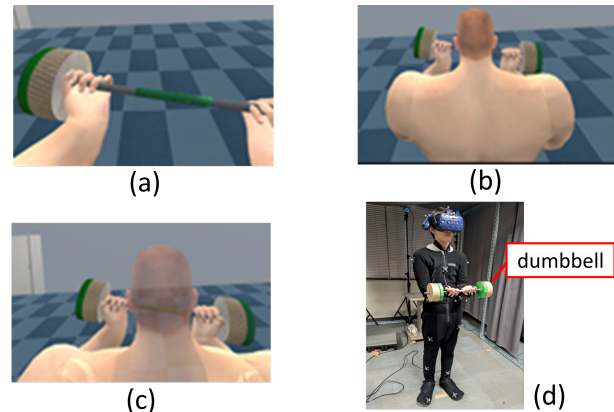


Figure 1: View from each perspective: (a) the 1st person perspective, (b) the 3rd person perspective, (c) the "1st+3rd" person perspective, (d) Participant during the experiment

Building on these findings, we previously proposed a novel viewpoint termed the "1st+3rd" person perspective which combines the advantages of the 1st and 3rd person perspectives by incorporating semi-transparency into part of the avatar's body. In this study, we design a new weight comparison task to investigate the "1st+3rd" person perspective in more detail, and the influence of the impression of the avatar's strength on the weight illusion.

2 The "1st+3rd" Person Perspective

The "1st+3rd" person perspective ("1+3" PP) is defined as a viewpoint in which sees through a semi-transparent avatar with a region around the head made partially transparent. The viewpoint is shifted backward from the avatar's head to the extent that it does not impair the sense of agency, while being positioned so that the user can clearly see the avatar's body in a way that enhances the sense of ownership. "1+3" PP is individually adjusted for each user.

Figure 1 shows the views in the 1st person perspective (1PP), the 3rd person perspective (3PP), and "1+3" PP. "1+3" PP combines the advantage of 3PP, where the avatar's body can be fully observed, with the advantage of 1PP, where information in front of the body can be obtained by making parts of the user's body semi-transparent during movement.

3 Experimental Overview

This experiment investigates the effects of 1PP, 3PP, and "1+3" PP on the weight illusion, by examining differences in weight perception when lifting the same weight from each perspective. Additionally, we explore how the sense of ownership, the sense of agency, and the impression of the avatar's strength correlate with weight illusion.

3.1 Viewpoint Selection Task

In the viewpoint selection task, participants select the viewpoint positions for 3PP and "1+3" PP used in the weight evaluation task. They are instructed to make their selections based on how easily they can embody the avatar and how strongly they can feel the avatar's strength when lifting objects. Additionally, for "1+3" PP, participants also select the transparency level and transparency area. Participants then evaluate the sense of ownership, the sense of agency, and the impression of the avatar's strength under 1PP, 3PP, and the "1+3" PP. After freely moving their bodies to sufficiently embody the avatar in each perspective, they answer the following questions (Q1–Q3). Q1 and Q2 are rated on a 7-point Likert scale ranging from -3 ("not at all") to +3 ("very strongly").

Q1: Did you feel as if the avatar's body was your own body?

Q2: Did you feel that your movements matched those of the avatar?

Q3: In this state, how many times the maximum weight you can normally lift by yourself do you think you could lift? (to the first decimal place)

3.2 Weight Evaluation Task

In the weight evaluation task, participants lift a dumbbell and evaluate its weight under 1PP, 3PP, and "1+3" PP. The evaluation of weight is conducted by comparing it with the impression of the weight of an object in a reference image that participants have selected in advance. The reference image for comparison is chosen from a set of 12 images (objects ranging from 0.5 kg to 5.0 kg), and is the image containing the object whose weight is perceived as closest to that of the dumbbell after participants actually lift the dumbbell in the real environment.

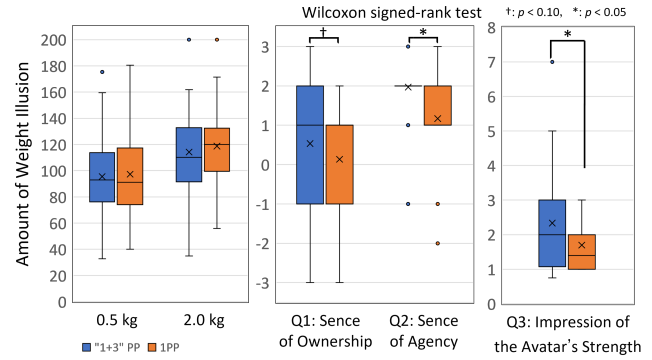
The procedure is as follows:

- (1) In the real world, participants lift the dumbbell and choose a reference image. They then report the weight of the object in the reference image, assuming the weight ratio of the dumbbell lifted in the real environment is 1.0.
- (2) Wearing an HMD, they move their body until they sufficiently embody the avatar.
- (3) They lift the dumbbell and evaluate its weight, assuming the weight of the object in the reference image is 100.
- (4) Participants removes the HMD and stack paper cups.

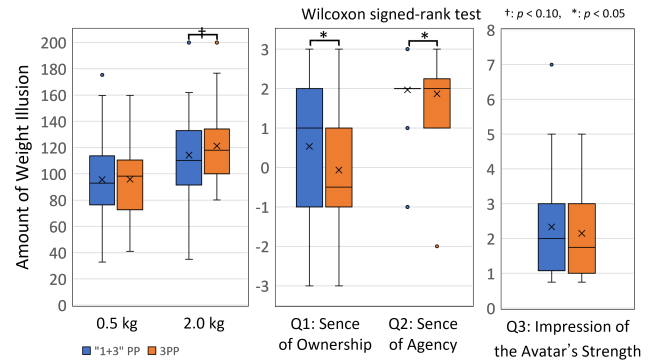
Here, the weight of the object in the reference image from (1) is used to correct the amount of weight illusion, in order to reduce the influence of cases where the selected reference image greatly differs from the actual weight of the dumbbell. In (4), participants stack paper cups, which are sufficiently light, to reset the perception of the dumbbell's weight and avoid carry-over effects into the next trial. (2) to (4) constitute one trial. Participants complete one trial each under 1PP, 3PP, and "1+3" PP, for a total of three trials. The order of perspectives is randomized. In addition, (1) through (2)–(4) together constitute one set, and the procedure is repeated for three sets. The first set is a practice trial using a 0.5 kg dumbbell. In the remaining two sets, 0.5 kg and 2.0 kg dumbbells are used, with the order randomized for each participant.

4 Results and Discussion

The participants were 30 individuals in their 20s. Figure 2 shows the graphs comparing "1+3" PP with 1PP and 3PP as the baselines.



(a) The "1st+3rd" person perspective and the 1st person perspective



(b) The "1st+3rd" person perspective and the 3rd person perspective

Figure 2: Comparison results (The vertical axis for Q1, Q2 is the 7-point Likert scale and that for Q3 represents the multiple.)

The amount of weight illusion in (3) is the perceived weight of the dumbbell when the weight of the object in the reference image is assumed to be 100. In the figures, the amount of weight illusion is corrected by multiplying it with the weight ratio of the object in the reference image provided by each participant.

From the present results, "1+3" PP was found to provide a significantly higher sense of ownership than 3PP and a significantly higher sense of agency than 1PP, indicating that "1+3" PP combines the advantages of both 1PP and 3PP. Furthermore, "1+3" PP yielded a significantly higher impression of the avatar's strength than 1PP, demonstrating its effectiveness as a method to enhance the impression of the avatar's strength. Regarding the amount of weight illusion, a tendency for "1+3" PP to perceive objects as lighter than 3PP was observed only at 2.0 kg, suggesting that heavier objects may elicit a stronger illusion.

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